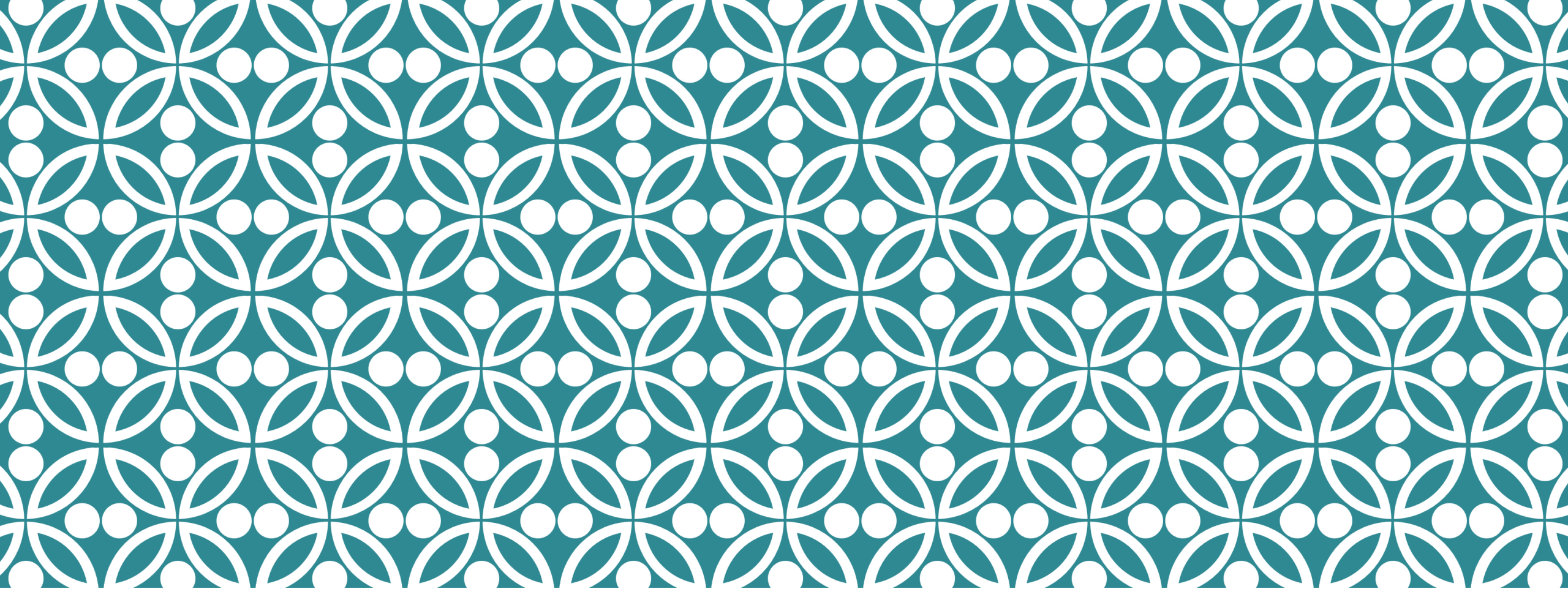




# SESSION 7: TEXT AS DATA

DR. SOFIA GIL-CLAVEL

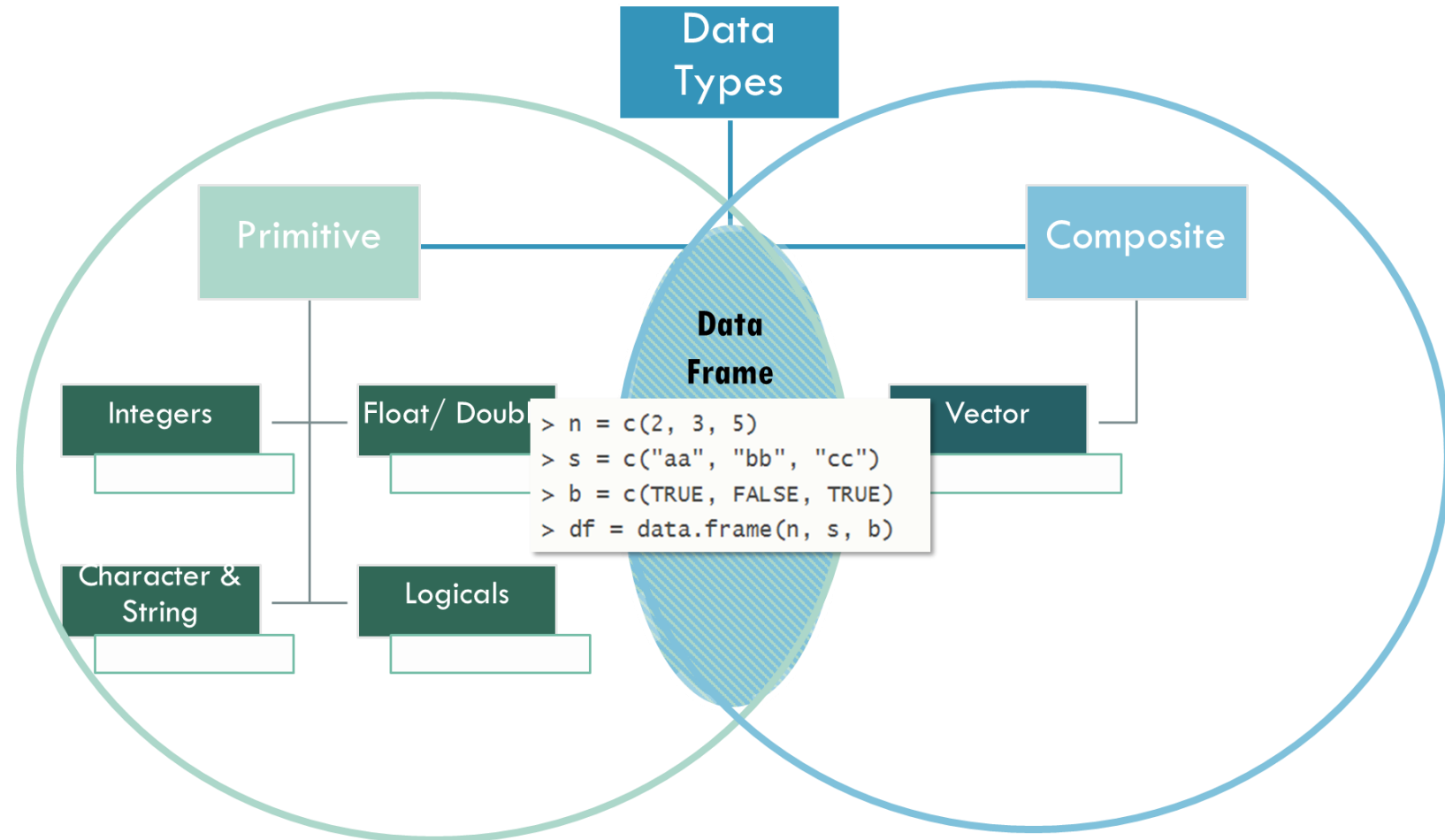
❖ The basics of handling text in R.

# 0. SESSION 1-6: RECAP AND CONNECTION

---

- R and RStudio
- Data types
- Data frames

## Introduction to R and Data frames



## Introduction to R and Data frames

## Using packages: Tidyverse

- Packages
- Tidy logic
- Dplyr

1.



```
> iris
# A tibble: 150 × 5
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
    <dbl>         <dbl>         <dbl>         <dbl>   <fct>
1     5.1         3.5         1.4         0.2   setosa
2     4.9         3         1.4         0.2   setosa
3     4.7         3.2         1.3         0.2   setosa
4     4.6         3.1         1.5         0.2   setosa
5      5          3.6         1.4         0.2   setosa
6     5.4         3.9         1.7         0.4   setosa
7     4.6         3.4         1.4         0.3   setosa
8      5          3.4         1.5         0.2   setosa
9     4.4         2.9         1.4         0.2   setosa
10    4.9         3.1         1.5         0.1   setosa
#> 140 more rows
#> print(n = ...) to see more rows
```



2.



### Basic piping

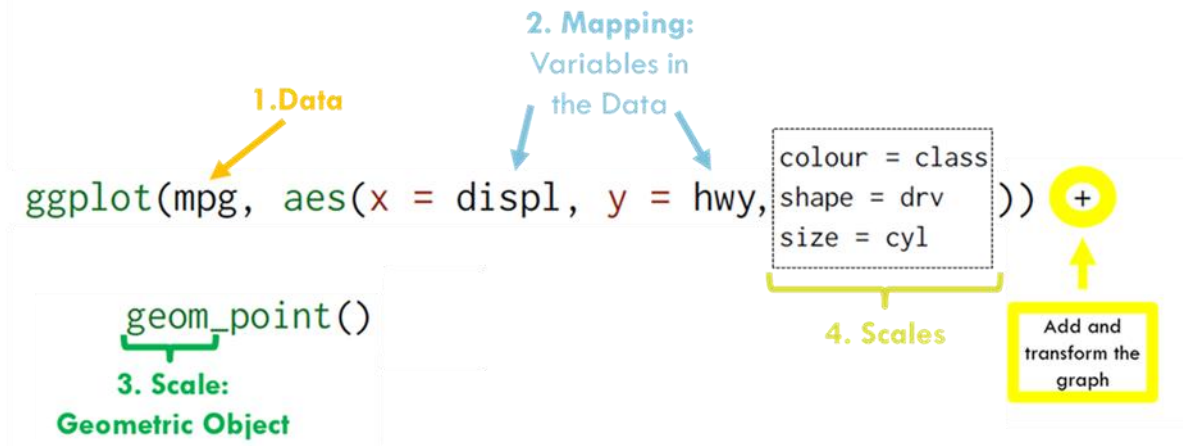
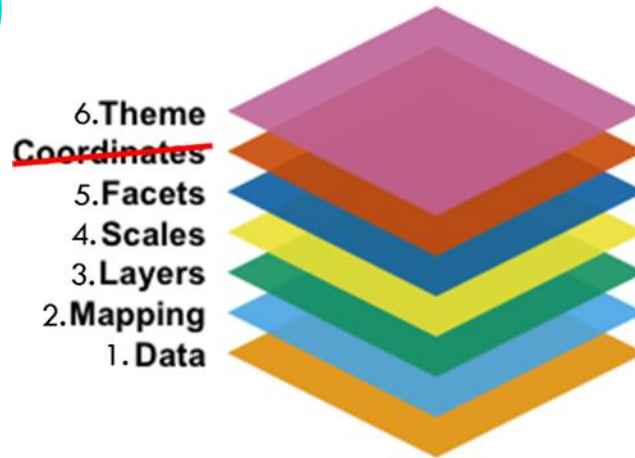
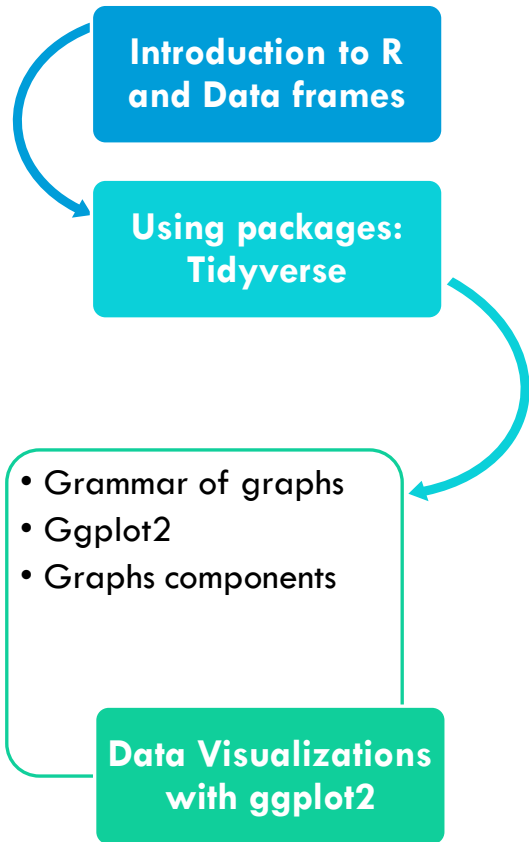
- `x %>% f` is equivalent to `f(x)`
- `x %>% f(y)` is equivalent to `f(x, y)`
- `x %>% f %>% g %>% h` is equivalent to `h(g(f(x)))`

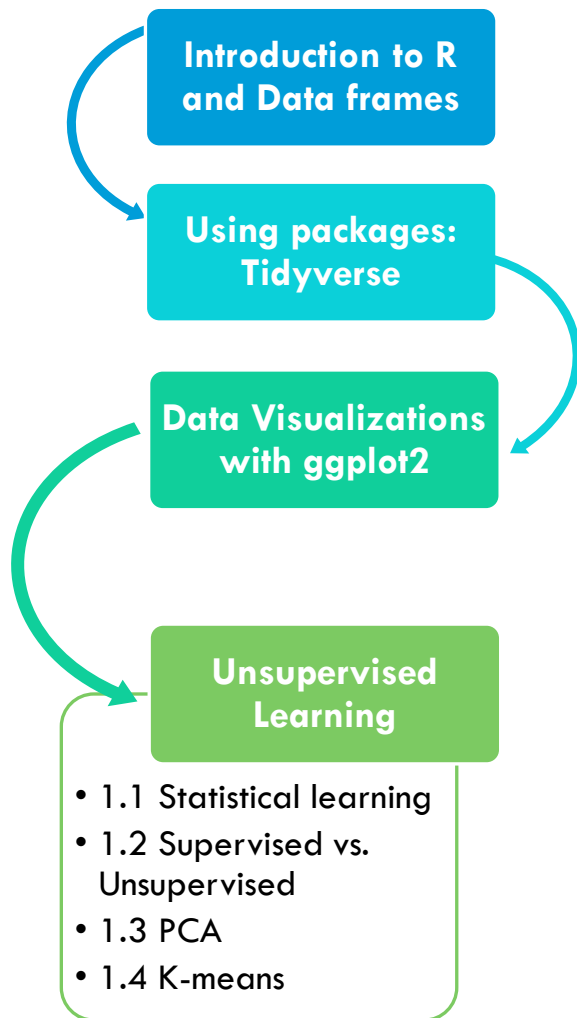
3.



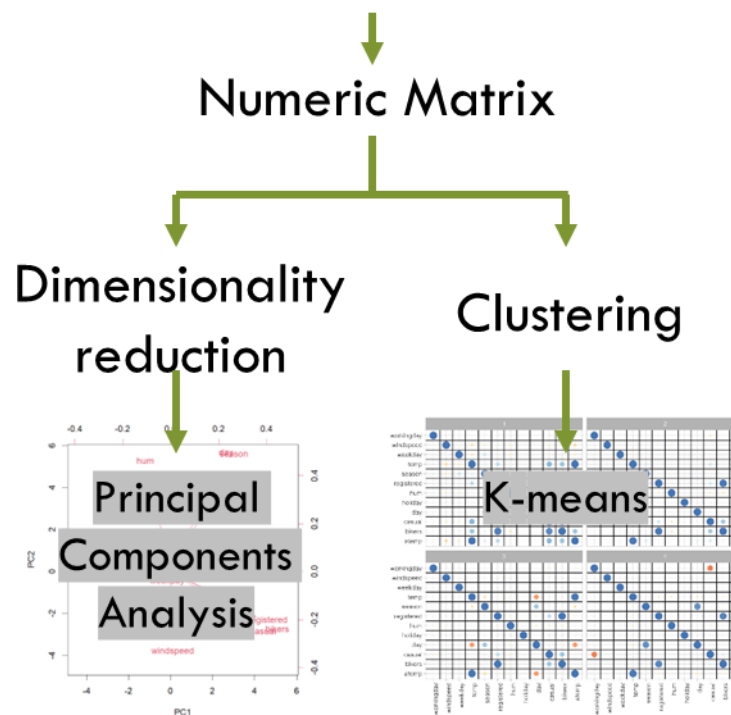
dplyr is a grammar of data manipulation, providing a consistent set of verbs that help you solve the most common data manipulation challenges:

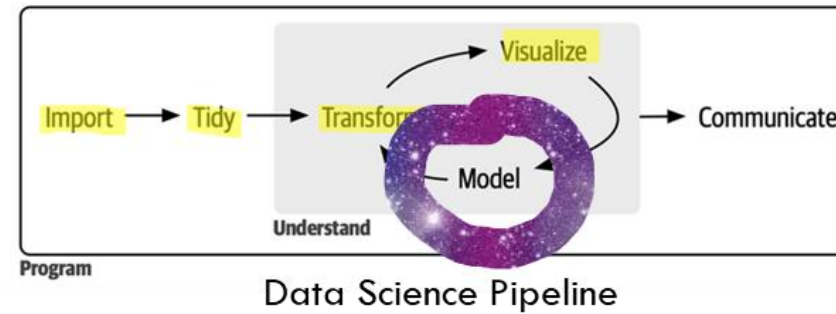
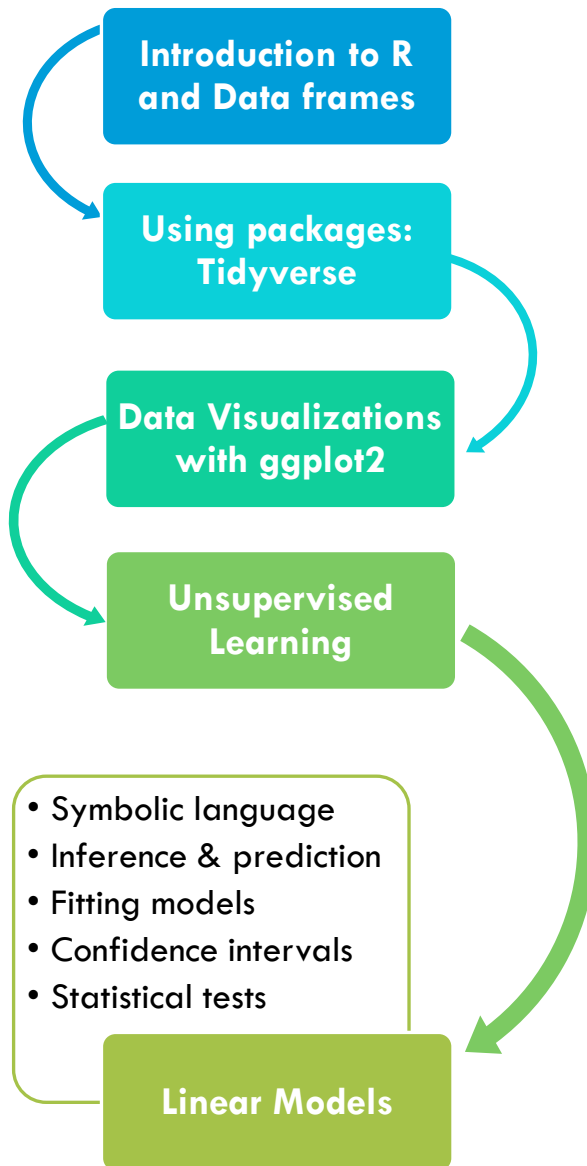
- `mutate()` adds new variables that are functions of existing variables
- `select()` picks variables based on their names.
- `filter()` picks cases based on their values.
- `summarise()` reduces multiple values down to a single summary.
- `arrange()` changes the ordering of the rows.





  $\sim X1_i + X2_i + X3_i$





stats-package {stats} R Documentation

The R Stats Package

Description

R statistical functions

Details

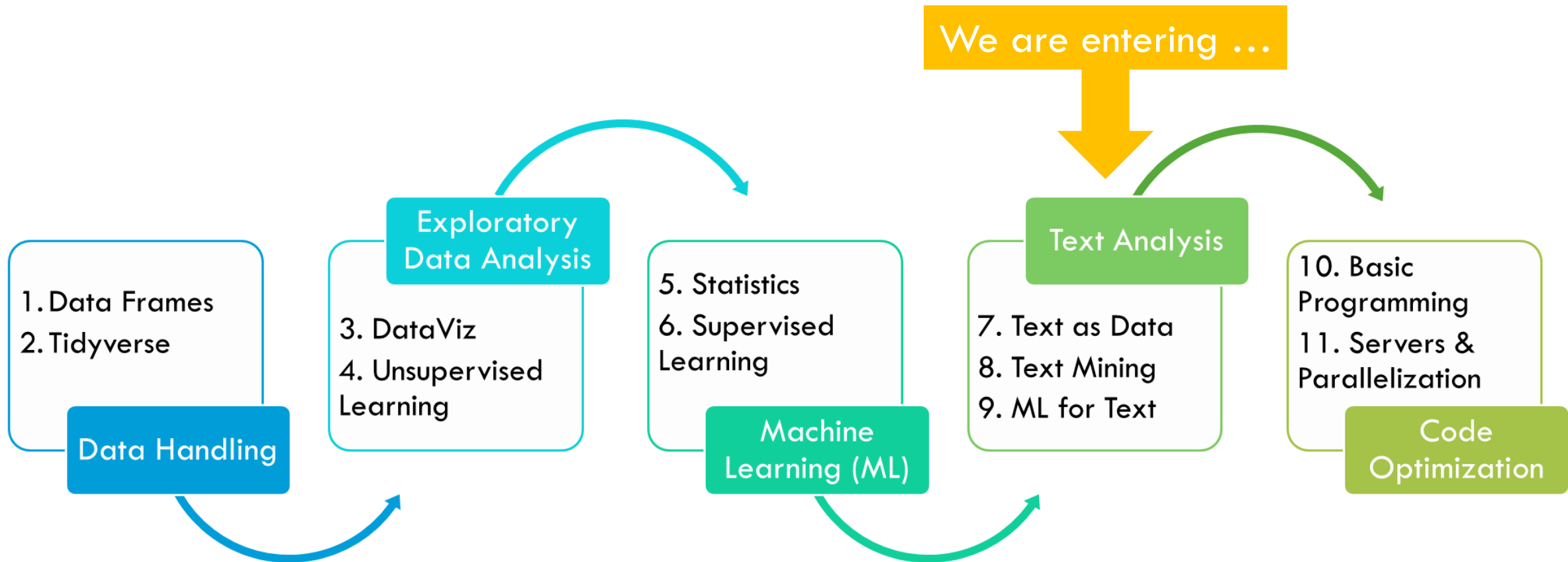
This package contains functions for statistical calculations and random number generation.

For a complete list of functions, use `library(help = "stats")`.

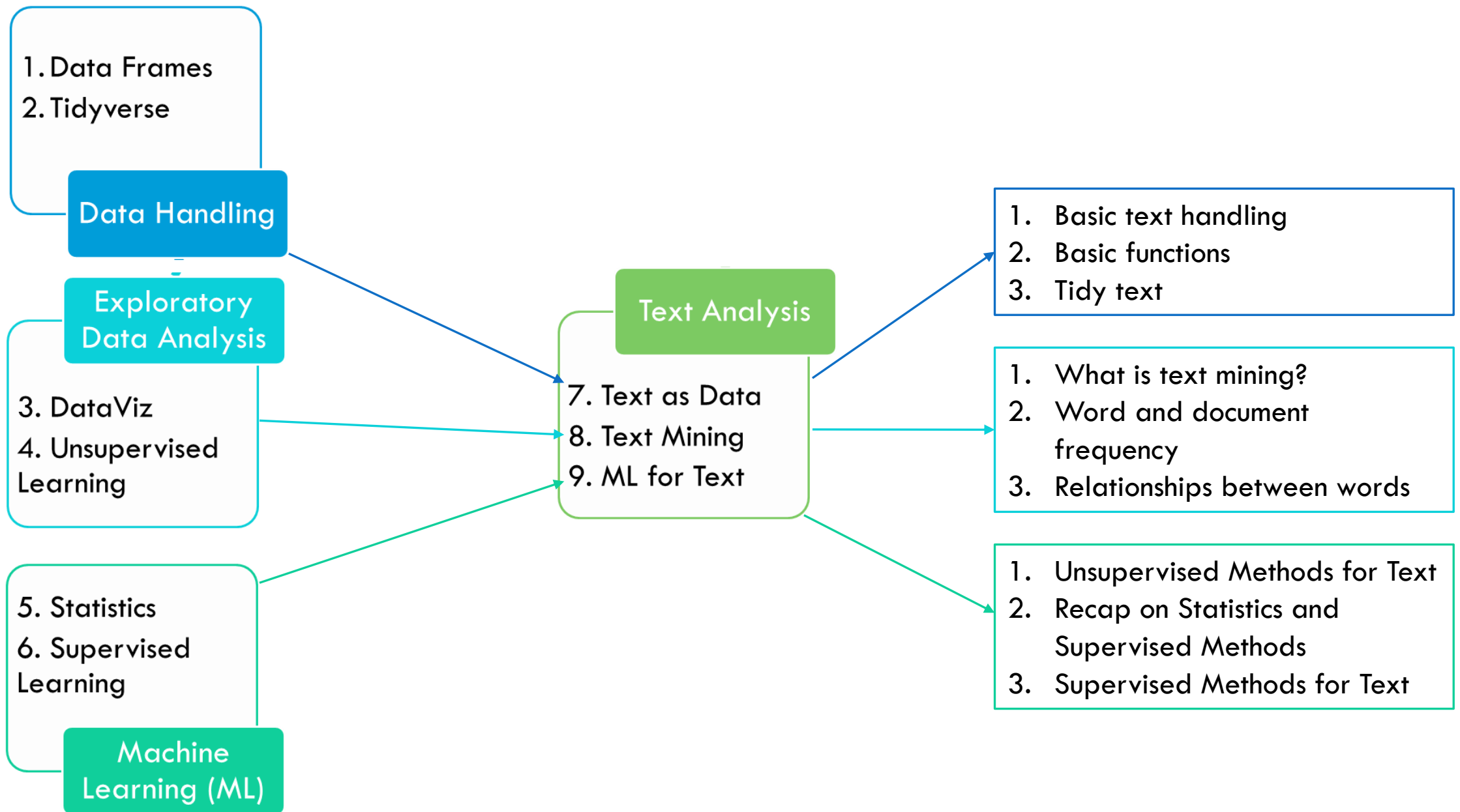
Author(s)

R Core Team and contributors worldwide

Maintainer: R Core Team [R-core@r-project.org](mailto:R-core@r-project.org)







# 1. THE BASICS OF HANDLING TEXT IN R

---

1. Basic function with Stringr
2. Basic text handling
3. Tidy text
4. Basic DataViz

# 1. BASIC FUNCTIONS

# STRINGR FOR BASIC TEXT HANDLING

R has its own functions to handle text, but it is not that convenient because the names are neither intuitive nor standardized. We will solve those problems by using the package “stringr”.



Install and open the package in R.

# SOME STRINGR FUNCTIONS

## stringr

```
str_detect(string, pattern)
```

```
str_dup(string, times)
```

```
str_extract(string, pattern)
```

```
str_extract_all(string, pattern)
```

```
str_length(string)
```

```
str_locate(string, pattern)
```

```
str_locate_all(string, pattern)
```

```
str_match(string, pattern)
```

```
str_order(string)
```

```
str_replace(string, pattern, replacement)
```

```
str_replace_all(string, pattern, replacement)
```

## stringr

```
str_sort(string)
```

```
str_split(string, pattern)
```

```
str_sub(string, start, end)
```

```
str_subset(string, pattern)
```

```
str_to_lower(string)
```

```
str_to_title(string)
```

```
str_to_upper(string)
```

```
str_trim(string)
```

```
str_which(string, pattern)
```

```
str_wrap(string)
```

# EXERCISE 1.1

Using the stringr functions, answer the following questions:

- How many characters are in the text?

Hint: Use the function `str_length()`

- Is the space also considered a character?

Hint: Use the function `str_detect()`

- How would you get the number of characters without considering the spaces?

Hint: Use the function `str_replace_all()`

- How would you turn all the characters into lowercase?

Hint: Use the function `str_to_lower()`

# EXERCISE 1.2

➤ Let's turn the data into a tibble, so that we can use the tidyverse package.

1. Split the string by word and save the result in the variable TXT2.

Hint: `str_split()`

2. Save the vector in a tibble, remember that the result comes as a list.

Hint: Extract the vector doing `TXT2[[1]]`.

3. Change the name of the variable to “word”.

Hint: Use the function `colnames()`.

4. Use the tidyverse packages to count the number of times each word appeared in the text.

Hint: Use the functions `group_by()`, `summarise()`, and `n()`.

5. Arrange the tibble by word frequency in descending order.

Hint: Use the functions `arrange()` and `desc()`.

## 2. BASIC TEXT HANDLING

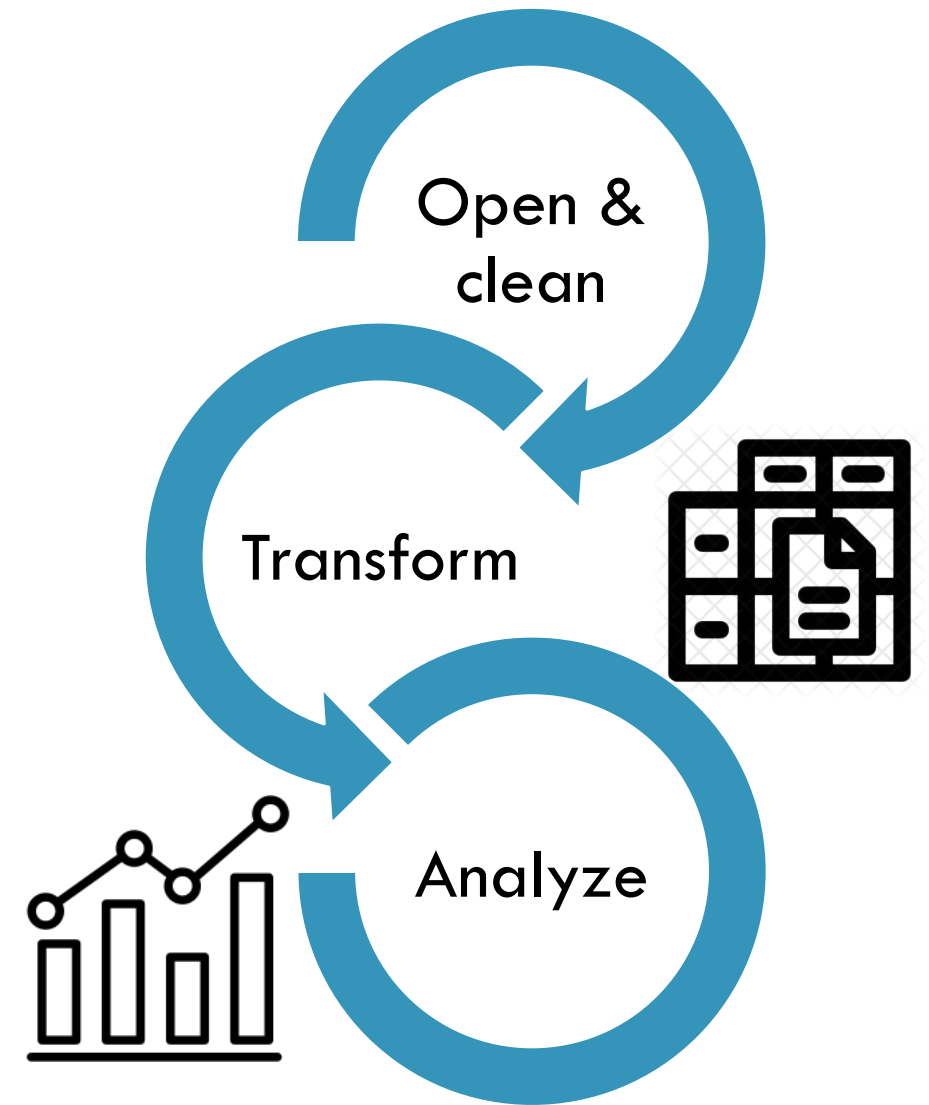


# THE PIPELINE

Today, I went to the café in the city center to meet my friends. We sat at a table near the fenêtre, enjoying the view of the bustling streets. The ambiente inside was filled with the profumo of freshly baked croissants and the sonido of laughter. It felt so acogliente and charming, as if time had slowed down just for us. We chatted about everything from life to our latest aventuras in the mountains. One of my friends, Mário, told us about his travesía through the desierto last year. We all agreed that viajar is one of the greatest joys in life.

The weather was perfect, with a brisa coming from the sea, and the sun shining brightly. We couldn't resist sitting outside on the terrace, where we could feel the fresh air on our skin. The streets around us were filled with people walking with sonrientes faces, enjoying the luz of the day. It was the kind of day where everything seemed to pause, and we embraced the moment fully.

As we sipped our cappuccini, we talked about our plans for the summer. Giulia suggested we take a trip to the montaña to escape the heat. Her idea sounded fantastico, and we all agreed to start planning it. We laughed about how we caminariamos all day, lost in the beauty of nature. After a while, the conversation shifted to películas and funny moments from the past, and we found ourselves reminiscing about the most ridículo things that had happened over the years.

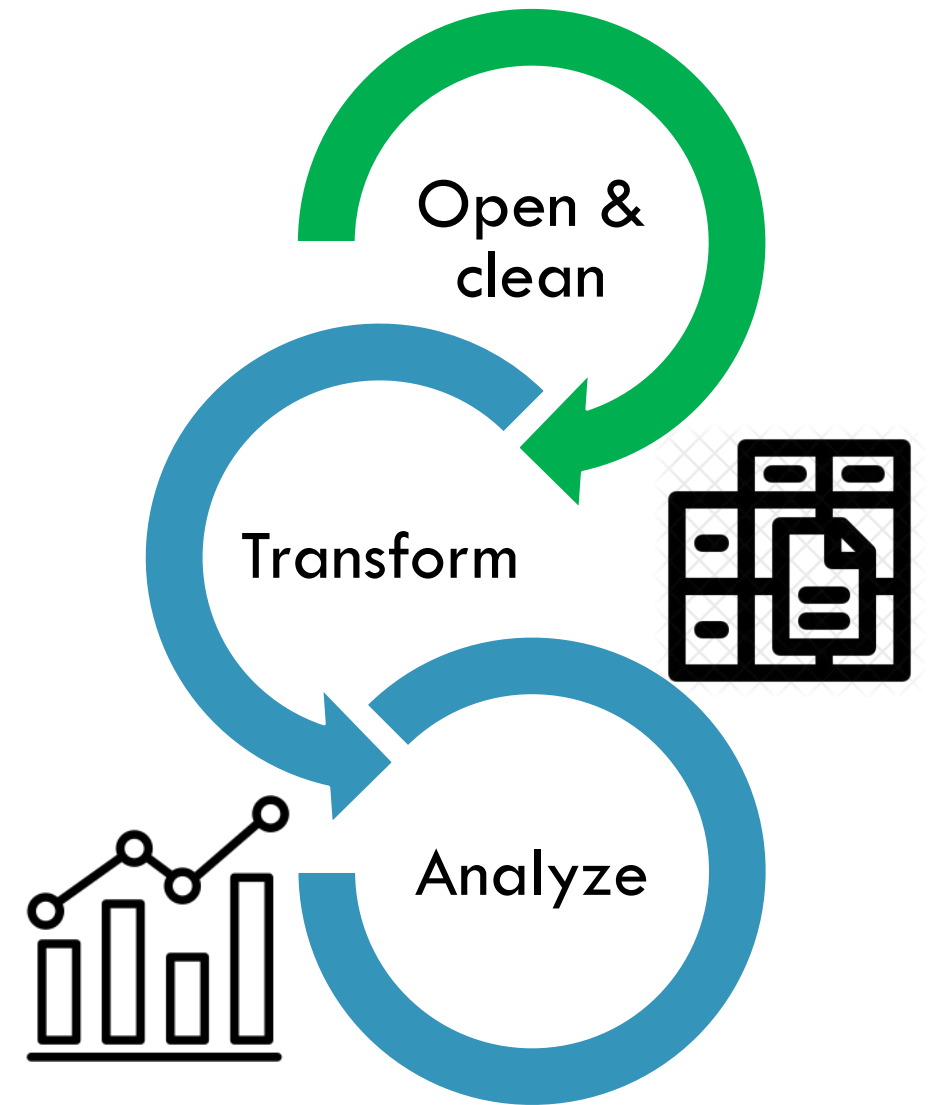


# THE PIPELINE

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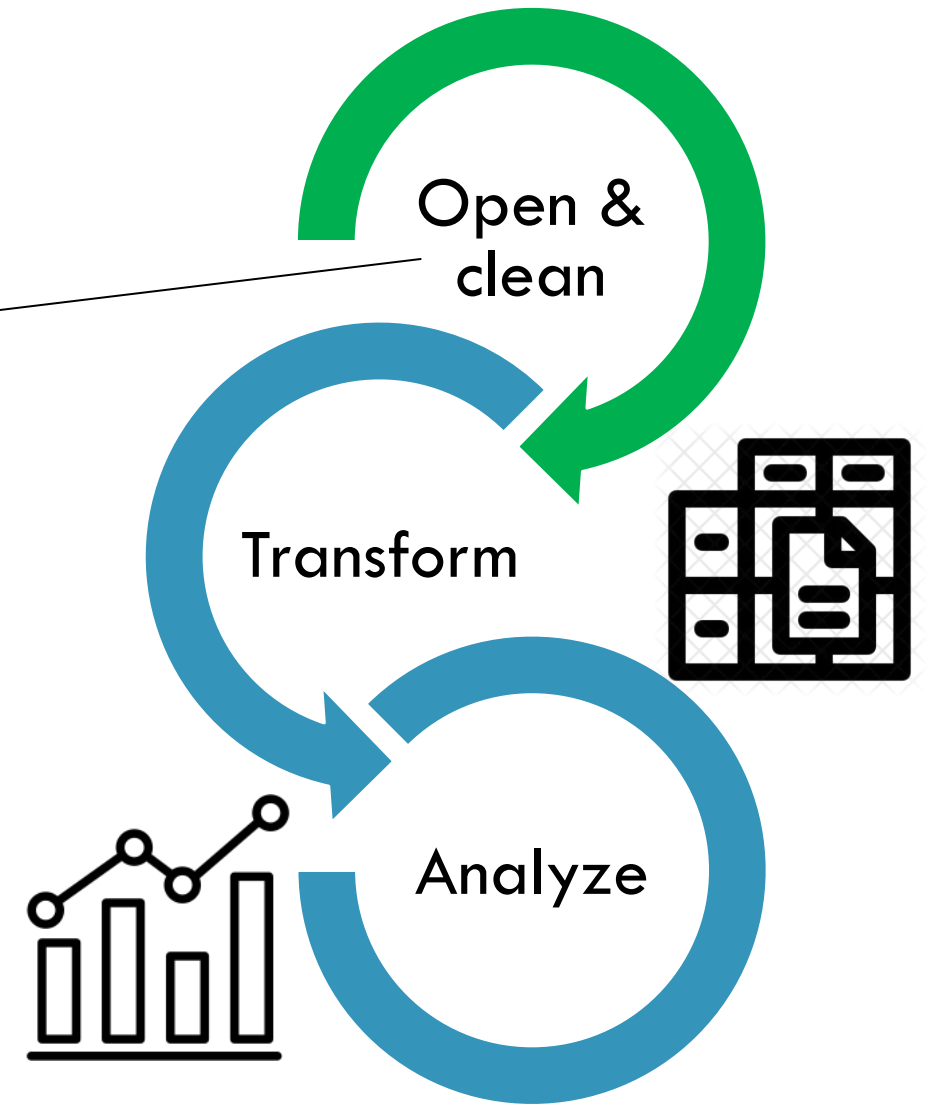
# THE PIPELINE

Today, I went to the café in the city center to meet my friends. We sat at a table near the fenêtre, enjoying the view of the bustling streets. The ambiente inside was filled with the profumo of freshly baked croissants and the sonido of laughter. It felt so acogliente and charming, as if time had slowed down just for us. We chatted about everything from life to our latest aventuras in the mountains. One of my friends, Mário, told us about his travesía through the desierto last year. We all agreed that viajar is one of the greatest joys in life.

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Let's open the text "Example\_Text.txt"  
What problems do you encounter?



# Types of Character Data

## El código ASCII

sigla en inglés de American Standard Code for Information Interchange  
( Código Estadounidense Estándar para el Intercambio de Información )

| Caracteres de control ASCII |     |               |                       |
|-----------------------------|-----|---------------|-----------------------|
| DEC                         | HEX | Simbolo ASCII |                       |
| 00                          | 00h | NULL          | (carácter nulo)       |
| 01                          | 01h | SOH           | (inicio encabezado)   |
| 02                          | 02h | STX           | (inicio texto)        |
| 03                          | 03h | ETX           | (fin de texto)        |
| 04                          | 04h | EOT           | (fin transmisión)     |
| 05                          | 05h | ENQ           | (enquiry)             |
| 06                          | 06h | ACK           | (acknowledgement)     |
| 07                          | 07h | BEL           | (timbre)              |
| 08                          | 08h | BS            | (retroceso)           |
| 09                          | 09h | HT            | (tab horizontal)      |
| 10                          | 0Ah | LF            | (salto de línea)      |
| 11                          | 0Bh | VT            | (tab vertical)        |
| 12                          | 0Ch | FF            | (form feed)           |
| 13                          | 0Dh | CR            | (retorno de carro)    |
| 14                          | 0Eh | SO            | (shift Out)           |
| 15                          | 0Fh | SI            | (shift In)            |
| 16                          | 10h | DLE           | (data link escape)    |
| 17                          | 11h | DC1           | (device control 1)    |
| 18                          | 12h | DC2           | (device control 2)    |
| 19                          | 13h | DC3           | (device control 3)    |
| 20                          | 14h | DC4           | (device control 4)    |
| 21                          | 15h | NAK           | (negative acknowle.)  |
| 22                          | 16h | SYN           | (synchronous idle)    |
| 23                          | 17h | ETB           | (end of trans. block) |
| 24                          | 18h | CAN           | (cancel)              |
| 25                          | 19h | EM            | (end of medium)       |
| 26                          | 1Ah | SUB           | (substitute)          |
| 27                          | 1Bh | ESC           | (escape)              |
| 28                          | 1Ch | FS            | (file separator)      |
| 29                          | 1Dh | GS            | (group separator)     |
| 30                          | 1Eh | RS            | (record separator)    |
| 31                          | 1Fh | US            | (unit separator)      |
| 7                           | 20h | DEL           | (delete)              |

| Caracteres ASCII imprimibles |     |         |     |     |         |     |     |         |     |     |         |
|------------------------------|-----|---------|-----|-----|---------|-----|-----|---------|-----|-----|---------|
| DEC                          | HEX | Simbolo | DEC | HEX | Simbolo | DEC | HEX | Simbolo | DEC | HEX | Simbolo |
| 32                           | 20h | espacio | 64  | 40h | @       | 96  | 60h | `       |     |     |         |
| 33                           | 21h | !       | 65  | 41h | A       | 97  | 61h | a       |     |     |         |
| 34                           | 22h | "       | 66  | 42h | B       | 98  | 62h | b       |     |     |         |
| 35                           | 23h | #       | 67  | 43h | C       | 99  | 63h | c       |     |     |         |
| 36                           | 24h | \$      | 68  | 44h | D       | 100 | 64h | d       |     |     |         |
| 37                           | 25h | %       | 69  | 45h | E       | 101 | 65h | e       |     |     |         |
| 38                           | 26h | &       | 70  | 46h | F       | 102 | 66h | f       |     |     |         |
| 39                           | 27h | '       | 71  | 47h | G       | 103 | 67h | g       |     |     |         |
| 40                           | 28h | (       | 72  | 48h | H       | 104 | 68h | h       |     |     |         |
| 41                           | 29h | )       | 73  | 49h | I       | 105 | 69h | i       |     |     |         |
| 42                           | 2Ah | *       | 74  | 4Ah | J       | 106 | 6Ah | j       |     |     |         |
| 43                           | 2Bh | +       | 75  | 4Bh | K       | 107 | 6Bh | k       |     |     |         |
| 44                           | 2Ch | ,       | 76  | 4Ch | L       | 108 | 6Ch | l       |     |     |         |
| 45                           | 2Dh | -       | 77  | 4Dh | M       | 109 | 6Dh | m       |     |     |         |
| 46                           | 2Eh | .       | 78  | 4Eh | N       | 110 | 6Eh | n       |     |     |         |
| 47                           | 2Fh | /       | 79  | 4Fh | O       | 111 | 6Fh | o       |     |     |         |
| 48                           | 30h | 0       | 80  | 50h | P       | 112 | 70h | p       |     |     |         |
| 49                           | 31h | 1       | 81  | 51h | Q       | 113 | 71h | q       |     |     |         |
| 50                           | 32h | 2       | 82  | 52h | R       | 114 | 72h | r       |     |     |         |
| 51                           | 33h | 3       | 83  | 53h | S       | 115 | 73h | s       |     |     |         |
| 52                           | 34h | 4       | 84  | 54h | T       | 116 | 74h | t       |     |     |         |
| 53                           | 35h | 5       | 85  | 55h | U       | 117 | 75h | u       |     |     |         |
| 54                           | 36h | 6       | 86  | 56h | V       | 118 | 76h | v       |     |     |         |
| 55                           | 37h | 7       | 87  | 57h | W       | 119 | 77h | w       |     |     |         |
| 56                           | 38h | 8       | 88  | 58h | X       | 120 | 78h | x       |     |     |         |
| 57                           | 39h | 9       | 89  | 59h | Y       | 121 | 79h | y       |     |     |         |
| 58                           | 3Ah | :       | 90  | 5Ah | Z       | 122 | 7Ah | z       |     |     |         |
| 59                           | 3Bh | ;       | 91  | 5Bh | [       | 123 | 7Bh | {       |     |     |         |
| 60                           | 3Ch | <       | 92  | 5Ch | \       | 124 | 7Ch |         |     |     |         |
| 61                           | 3Dh | =       | 93  | 5Dh | ]       | 125 | 7Dh | }       |     |     |         |
| 62                           | 3Eh | >       | 94  | 5Eh | ^       | 126 | 7Eh | ~       |     |     |         |
| 63                           | 3Fh | ?       | 95  | 5Fh | -       |     |     |         |     |     |         |

elCodigoASCII.com.a

| ASCII extendido |     |         |     |     |         |     |     |         |     |     |         |     |     |         |     |     |         |
|-----------------|-----|---------|-----|-----|---------|-----|-----|---------|-----|-----|---------|-----|-----|---------|-----|-----|---------|
| DEC             | HEX | Simbolo | DEC | HEX | Simbolo | DEC | HEX | Simbolo | DEC | HEX | Simbolo | DEC | HEX | Simbolo | DEC | HEX | Simbolo |
| 128             | 80h | Ç       | 160 | A0h | á       | 192 | C0h | Ł       | 224 | E0h | Ó       |     |     |         |     |     |         |
| 129             | 81h | ü       | 161 | A1h | í       | 193 | C1h | ł       | 225 | E1h | ô       |     |     |         |     |     |         |
| 130             | 82h | é       | 162 | A2h | ó       | 194 | C2h | Ł       | 226 | E2h | Û       |     |     |         |     |     |         |
| 131             | 83h | â       | 163 | A3h | ú       | 195 | C3h | ł       | 227 | E3h | Ü       |     |     |         |     |     |         |
| 132             | 84h | ä       | 164 | A4h | ñ       | 196 | C4h | Ł       | 228 | E4h | ö       |     |     |         |     |     |         |
| 133             | 85h | à       | 165 | A5h | Ñ       | 197 | C5h | ł       | 229 | E5h | Õ       |     |     |         |     |     |         |
| 134             | 86h | á       | 166 | A6h | ª       | 198 | C6h | Ł       | 230 | E6h | µ       |     |     |         |     |     |         |
| 135             | 87h | ç       | 167 | A7h | º       | 199 | C7h | ł       | 231 | E7h | þ       |     |     |         |     |     |         |
| 136             | 88h | ê       | 168 | A8h | ¿       | 200 | C8h | Ł       | 232 | E8h | ß       |     |     |         |     |     |         |
| 137             | 89h | ë       | 169 | A9h | ®       | 201 | C9h | ł       | 233 | E9h | Ù       |     |     |         |     |     |         |
| 138             | 8Ah | è       | 170 | AAh | ¬       | 202 | CAh | Ł       | 234 | EAh | Ú       |     |     |         |     |     |         |
| 139             | 8Bh | ï       | 171 | ABh | ½       | 203 | CBh | ł       | 235 | EBh | Û       |     |     |         |     |     |         |
| 140             | 8Ch | î       | 172 | ACH | ¼       | 204 | CAh | Ł       | 236 | ECh | Ü       |     |     |         |     |     |         |
| 141             | 8Dh | ï       | 173 | ADh | »       | 205 | CDh | ł       | 237 | EDh | Ý       |     |     |         |     |     |         |
| 142             | 8Eh | Ä       | 174 | Aeh | «       | 206 | CEh | Ł       | 238 | EEh | ÿ       |     |     |         |     |     |         |
| 143             | 8Fh | Å       | 175 | AFh | »       | 207 | CFh | ł       | 239 | EFh | ·       |     |     |         |     |     |         |
| 144             | 90h | É       | 176 | B0h | ⋮       | 208 | D0h | ø       | 240 | F0h | ±       |     |     |         |     |     |         |
| 145             | 91h | æ       | 177 | B1h | ⋮       | 209 | D1h | Ð       | 241 | F1h | ±       |     |     |         |     |     |         |
| 146             | 92h | Æ       | 178 | B2h | ⋮       | 210 | D2h | É       | 242 | F2h | ±       |     |     |         |     |     |         |
| 147             | 93h | ø       | 179 | B3h | ⋮       | 211 | D3h | Ê       | 243 | F3h | ±       |     |     |         |     |     |         |
| 148             | 94h | ò       | 180 | B4h | ⋮       | 212 | D4h | Ë       | 244 | F4h | ±       |     |     |         |     |     |         |
| 149             | 95h | ó       | 181 | B5h | ⋮       | 213 | D5h | Ì       | 245 | F5h | ±       |     |     |         |     |     |         |
| 150             | 96h | û       | 182 | B6h | ⋮       | 214 | D6h | Í       | 246 | F6h | ±       |     |     |         |     |     |         |
| 151             | 97h | ü       | 183 | B7h | ⋮       | 215 | D7h | Î       | 247 | F7h | ±       |     |     |         |     |     |         |
| 152             | 98h | ÿ       | 184 | B8h | ⋮       | 216 | D8h | Ï       | 248 | F8h | ±       |     |     |         |     |     |         |
| 153             | 99h | Ö       | 185 | B9h | ⋮       | 217 | D9h | Ĵ       | 249 | F9h | ±       |     |     |         |     |     |         |
| 154             | 9Ah | Ü       | 186 | BAh | ⋮       | 218 | DAh | ⋮       | 250 | FAh | ±       |     |     |         |     |     |         |
| 155             | 9Bh | ø       | 187 | BBh | ⋮       | 219 | DBh | ⋮       | 251 | FBh | ±       |     |     |         |     |     |         |
| 156             | 9Ch | £       | 188 | BCh | ⋮       | 220 | DCh | ⋮       | 252 | FCh | ±       |     |     |         |     |     |         |
| 157             | 9Dh | ø       | 189 | BDh | ⋮       | 221 | DDh | ⋮       | 253 | FDh | ±       |     |     |         |     |     |         |
| 158             | 9Eh | x       | 190 | BEh | ⋮       | 222 | DEh | ⋮       | 254 | FEh | ±       |     |     |         |     |     |         |
| 159             | 9Fh | f       | 191 | BFh | ⋮       | 223 | DFh | ⋮       | 255 | FFh | ±       |     |     |         |     |     |         |

Characters that are not part of the standard ASCII and can be found in different encoding

# RELEVANCE OF USING THE RIGHT ENCODING

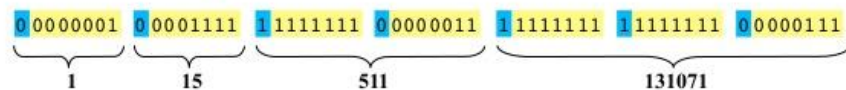
Strings of characters can be declared with different encodings, that is, depending on how the strings have been saved is the way in which the computer will process it.

## Byte-Aligned Variable-length Encodings

- Varint encoding:

- 7 bits per byte with continuation bit

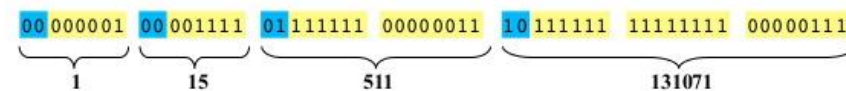
- Con: Decoding requires lots of branches/shifts/masks



- Idea: Encode byte length using 2 bits

- Better: fewer branches, shifts, and masks

- Con: Limited to 30-bit values, still some shifting to decode



Character repertoire

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| ´ | μ | ¶ | · | , | ´ |
| º | » | ¼ | ½ | ¾ | ¿ |
| À | Á | Â | Ã | Ä | Å |
| Æ | Ç | È | É | Ê | Ë |
| Ì | Í | Î | Ï | Ð | Ñ |

Á

U+00C1 LATIN CAPITAL LETTER A WITH ACUTE

General Character Properties

In Unicode since: 1.1  
Unicode category: Letter, Uppercase  
Canonical decomposition: U+0041 LATIN CAPITAL LETTER A + U+0301 COMBINING ACUTE ACCENT

Various Useful Representations

UTF-8: 0xC3 0x81  
UTF-16: 0x00C1

Octal escaped UTF-8: \303\201  
HTML decimal entity: &#193;  
Annotations and Cross References

Bytes

UTF-8

C3 81

character encodings

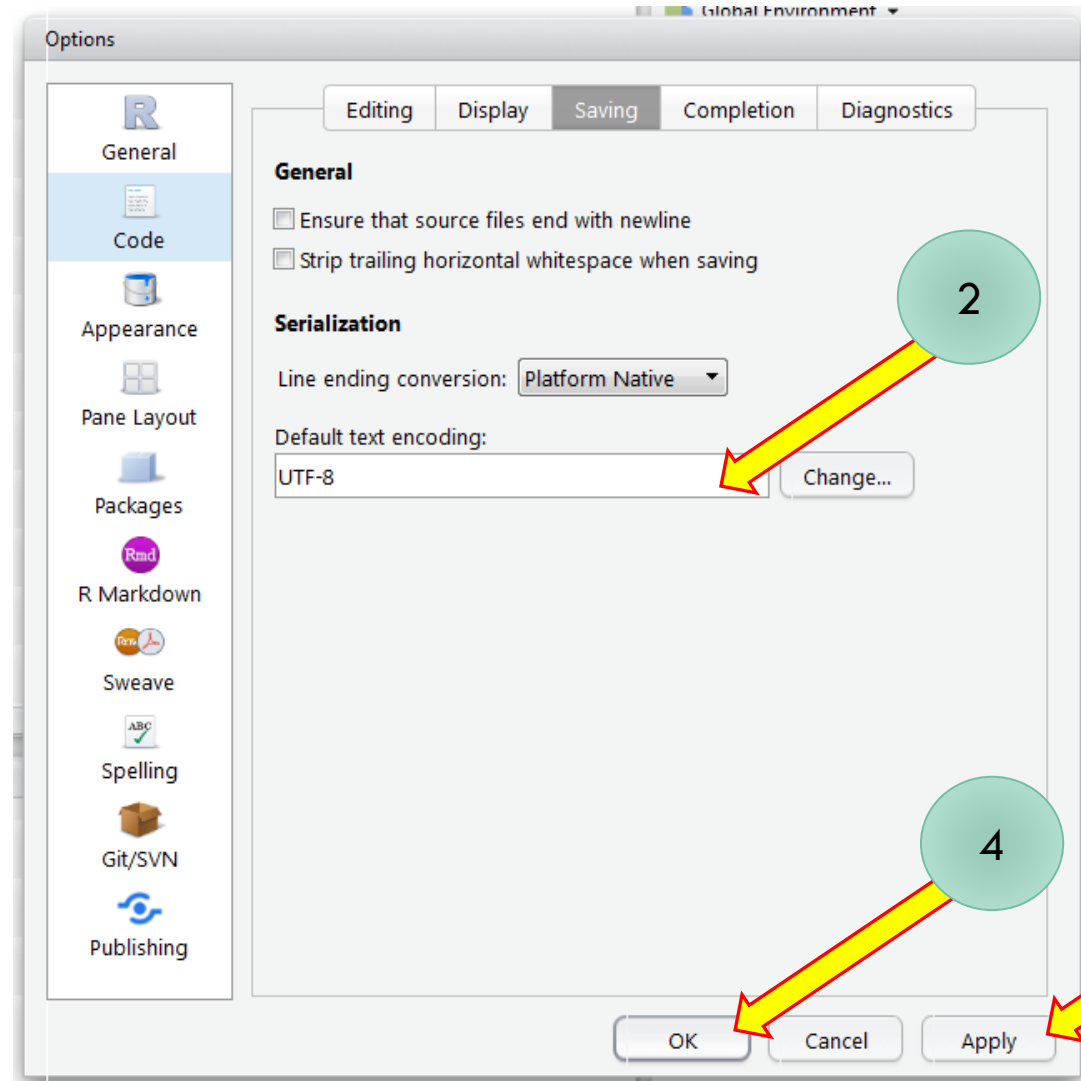
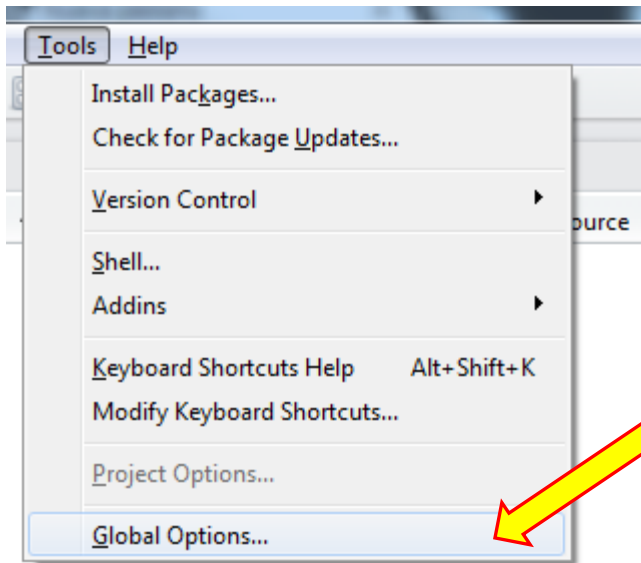
00 C1

UTF-16

That is why it is always important that before opening a document we are sure of the type of encoding that was used to save it.

```
> x <- "fa\xE7ile"
> Encoding(x)
[1] "latin1"
> xx <- iconv(x, "latin1", "UTF-8")
> Encoding(xx)
[1] "UTF-8"
> |
```

The most used type of encoding is Utf-8, which is why it is the type of encoding that will be handled here.



# 3. TIDY TEXT

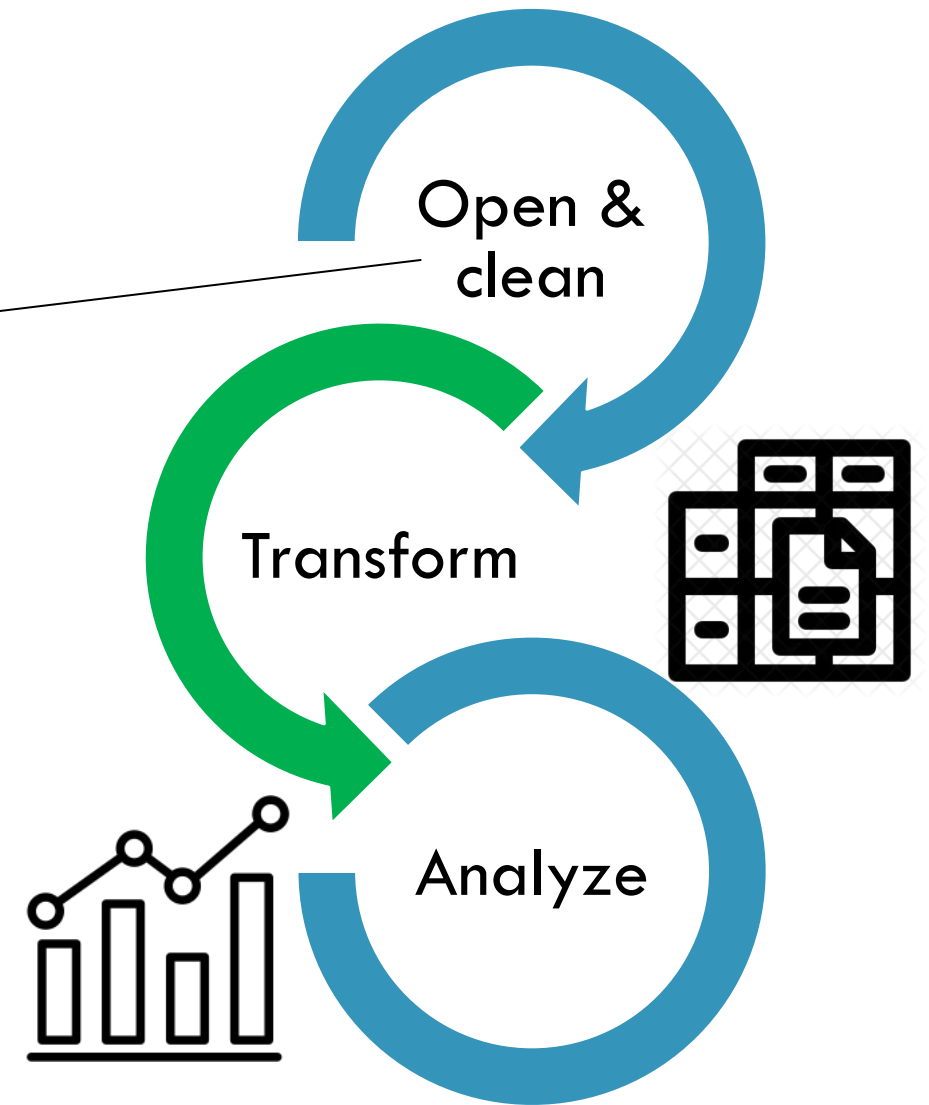


# THE PIPELINE

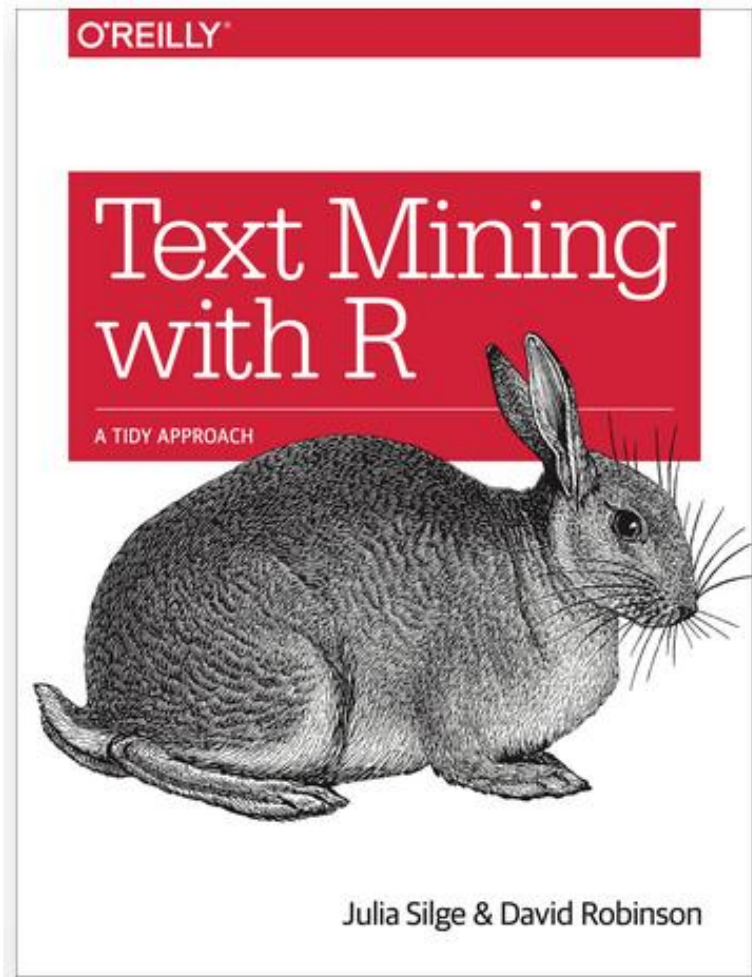
Today, I went to the café in the city center to meet my friends. We sat at a table near the fenêtre, enjoying the view of the bustling streets. The ambiente inside was filled with the profumo of freshly baked croissants and the sonido of laughter. It felt so acogliente and charming, as if time had slowed down just for us. We chatted about everything from life to our latest aventuras in the mountains. One of my friends, Mário, told us about his travesía through the desierto last year. We all agreed that viajar is one of the greatest joys in life.

The weather was perfect, with a brisa coming from the sea, and the sun shining brightly. We couldn't resist sitting outside on the terrace, where we could feel the fresh air on our skin. The streets around us were filled with people walking with sonrientes faces, enjoying the luz of the day. It was the kind of day where everything seemed to pause, and we embraced the moment fully.

As we sipped our cappuccini, we talked about our plans for the summer. Giulia suggested we take a trip to the montaña to escape the heat. Her idea sounded fantastico, and we all agreed to start planning it. We laughed about how we caminariamos all day, lost in the beauty of nature. After a while, the conversation shifted to películas and funny moments from the past, and we found ourselves reminiscing about the most ridículo things that had happened over the years.

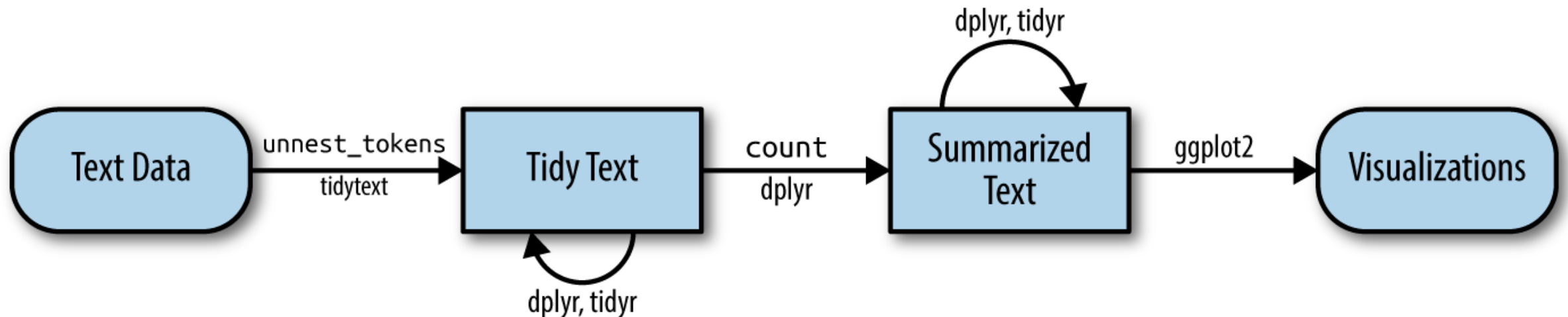


# ❖ TIDYTEXT: [HTTPS://WWW.TIDYTEXTMINING.COM/](https://www.tidytextmining.com/)



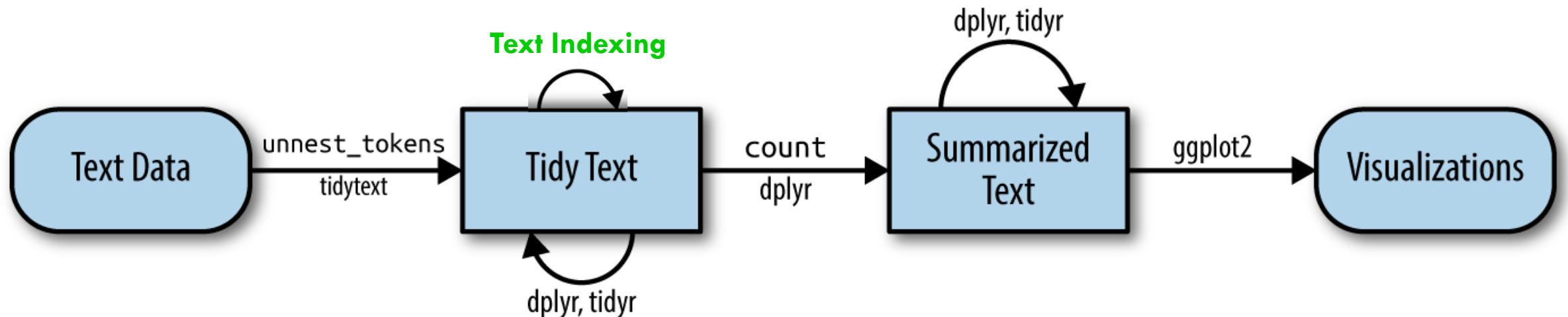
We developed the `tidytext` (Silge and Robinson 2016) R package because we were familiar with many methods for data wrangling and visualization, but couldn't easily apply these same methods to text. We found that using tidy data principles can make many text mining tasks easier, more effective, and consistent with tools already in wide use. Treating text as data frames of individual words allows us to manipulate, summarize, and visualize the characteristics of text easily and integrate natural language processing into effective workflows we were already using.

# FLOWCHART OF A TYPICAL TEXT ANALYSIS USING TIDY DATA PRINCIPLES.



Source: <https://www.tidytextmining.com/tidytext#tidytext>

# FLOWCHART OF A TYPICAL TEXT ANALYSIS USING TIDY DATA PRINCIPLES.

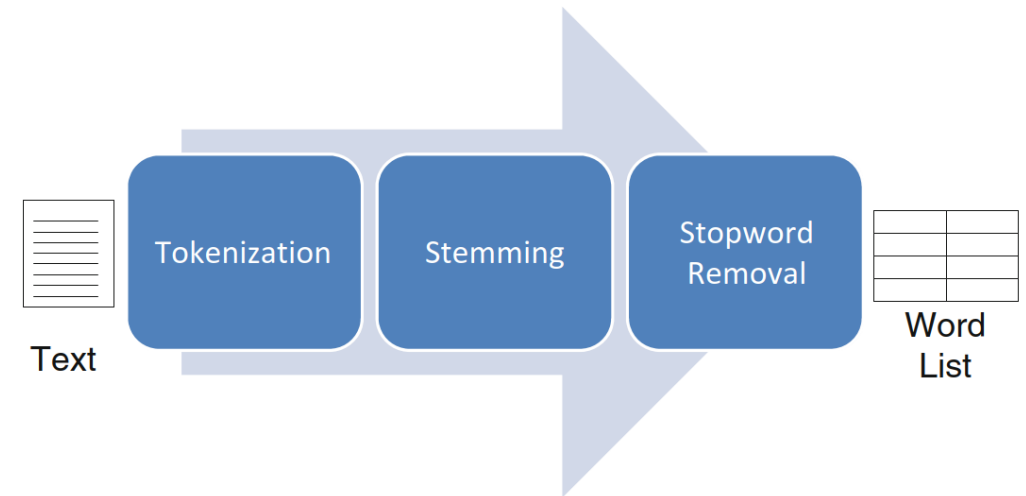


Source: <https://www.tidytextmining.com/tidytext#tidytext>

# OPTIONAL: TEXT INDEXING

➤ **Stemming:** When we deal with text, often documents contain different versions of one base word, often called a *stem*. What if we aren't interested in the difference between "trees" and "tree" and we want to treat both together? That idea is at the heart of stemming, the process of identifying the base word (or stem) for a data set of words.

➤ **Stopwords:** The grammatical words which are called stop words and used only for grammatical functions such as "a" or "the" have no meaning, so they are usually excluded in the text preprocessing and/or text indexing.



**Fig. 2.1** The three steps of text indexing

Source: <https://link.springer.com/book/10.1007/978-3-031-75976-5>

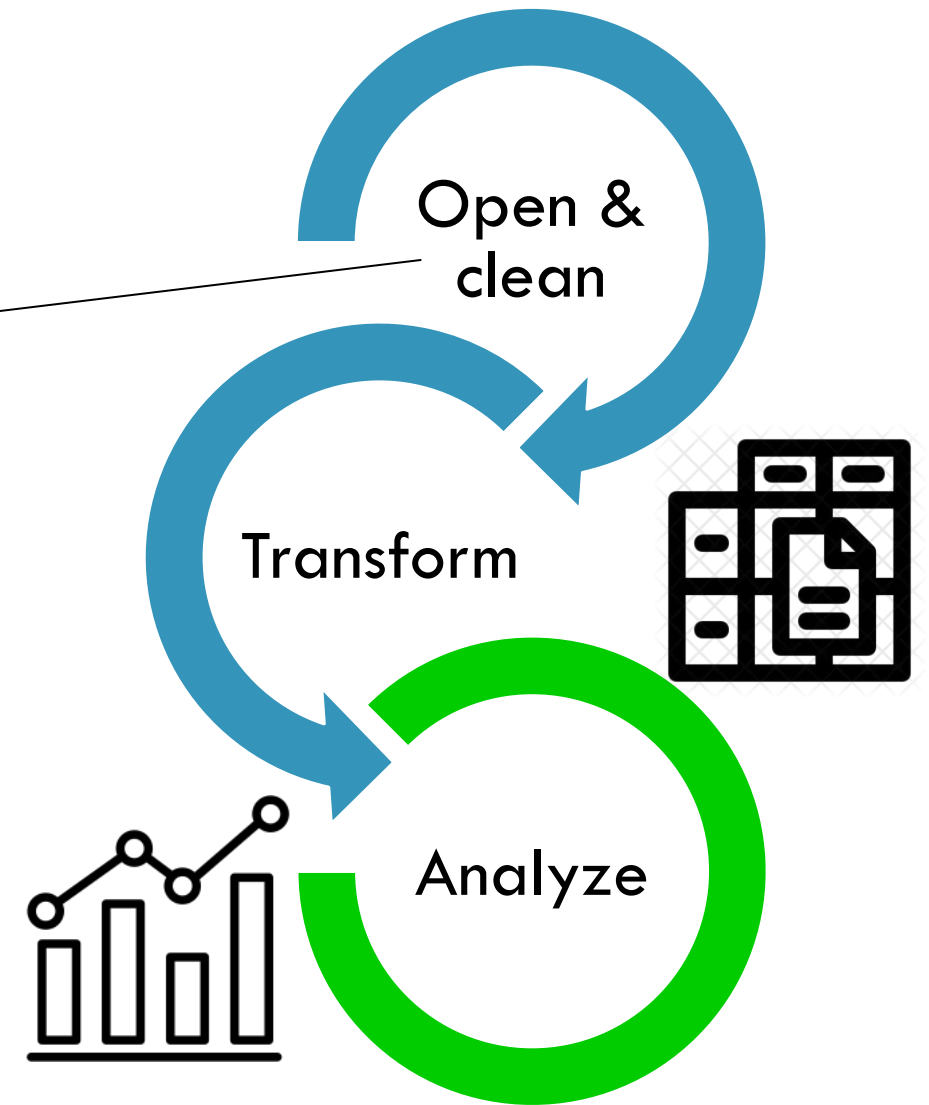
## 4. DATAVIZ

# THE PIPELINE

Today, I went to the café in the city center to meet my friends. We sat at a table near the fenêtre, enjoying the view of the bustling streets. The ambiente inside was filled with the profumo of freshly baked croissants and the sonido of laughter. It felt so acogliente and charming, as if time had slowed down just for us. We chatted about everything from life to our latest aventuras in the mountains. One of my friends, Mário, told us about his travesía through the desierto last year. We all agreed that viajar is one of the greatest joys in life.

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Join us!



<https://forms.office.com/e/8Bgd2YsasJ>

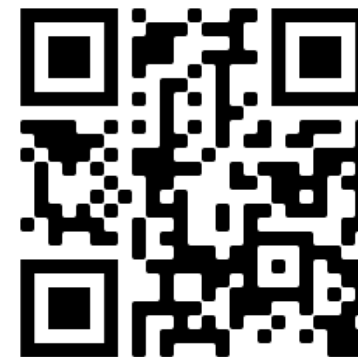
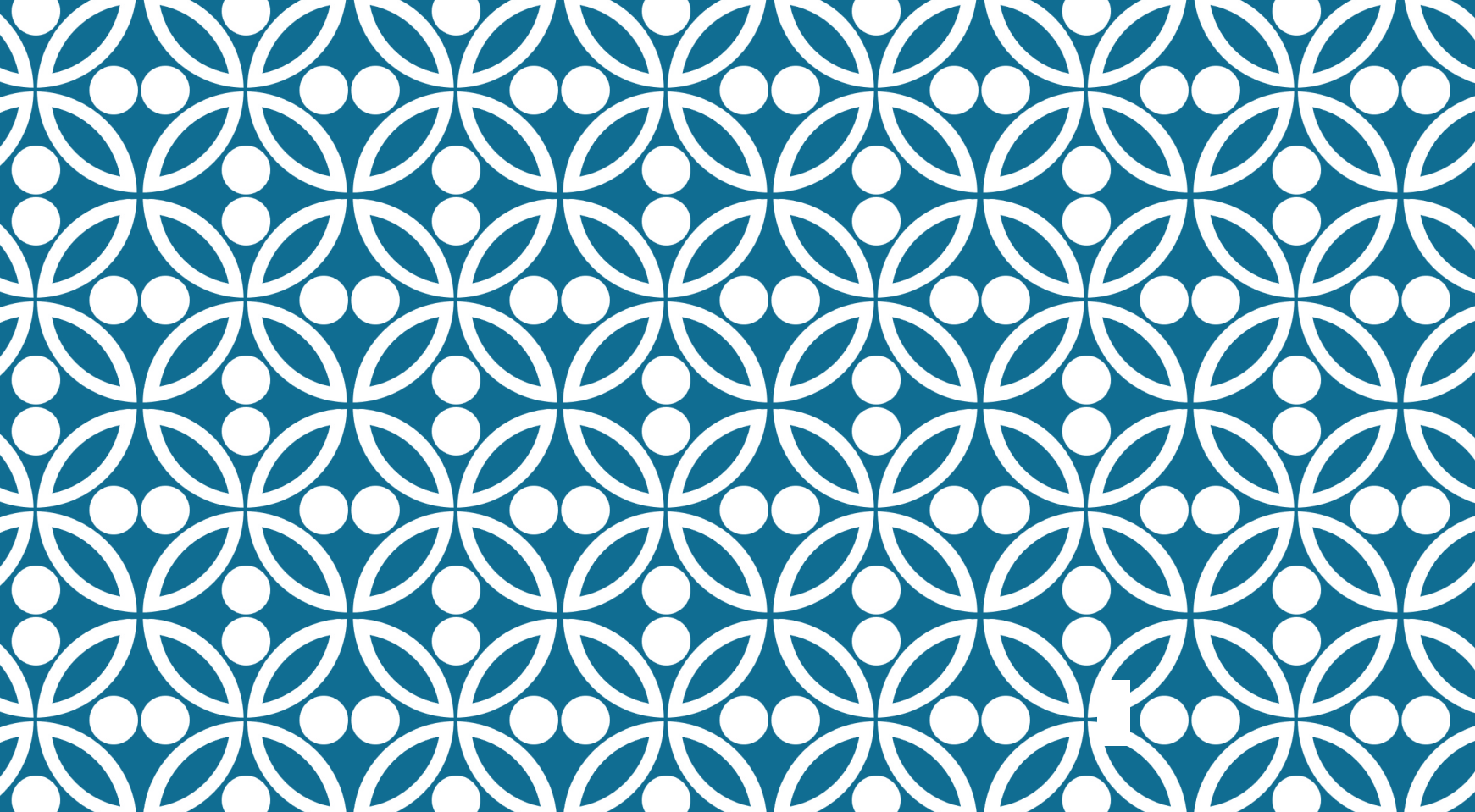
All about the lab:

<https://societal-analytics.nl/>

Contact us at:

[analytics-lab.fsw@vu.nl](mailto:analytics-lab.fsw@vu.nl)





<https://sofiag1l.github.io/>

# THANKS!

Dr. Sofia Gil-Clavel